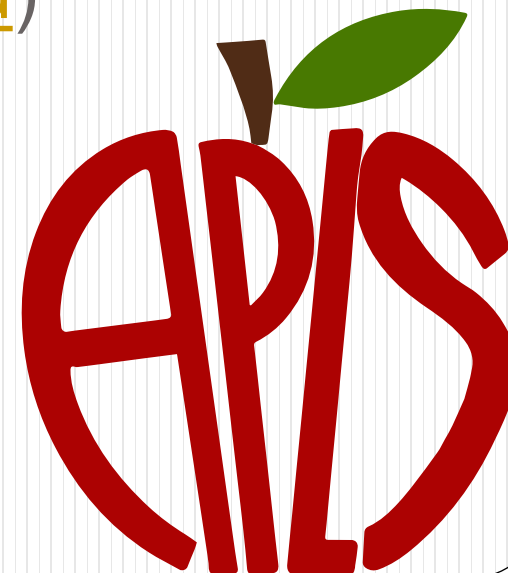


Pos[] Gazette and Gian[] Eagle: A replication of Guy (1980) with Pittsburgh English

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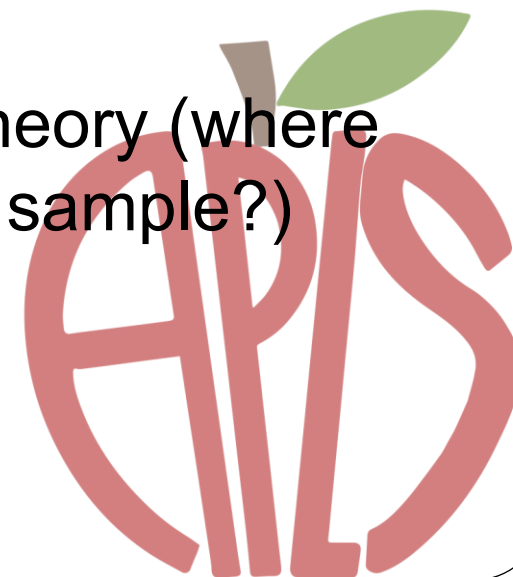
University of Pittsburgh



Speech Community as Shared Constraints hypothesis (SCSC)

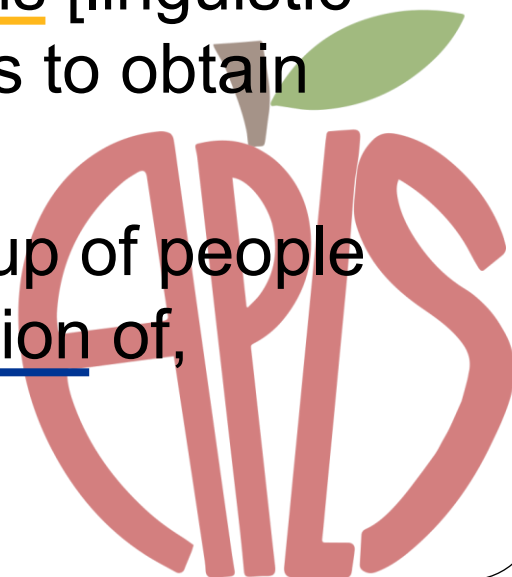
- Villarreal et al. (2021:245-246): SCSC is “the [mostly*] untested hypothesis that the existence of a shared constraint hierarchy...is one of the criteria that defines and delimits speech communities”
- SCSC in use:
 - Carmichael & Becker (2018): Very similar (r) constraint rankings for NYC vs. NoLA (but not BOUGHT raising), suggesting 19th-century diffusion from NYC to NoLA
- Defining the speech community is important for both theory (where does a lect begin and end?) and methods (who do we sample?)

* With apologies to Jon Forrest (2015)



Speech Community as Shared Constraints hypothesis (SCSC)

- Labov (2006/1966:82): NYC is “united by a common evaluation of the same variables which differentiate the speakers”
 - Measured via subjective evaluation tests, style-shifting toward standard
- But then along come variable rules and Varbrul
- Bickerton (1971) hyp: Observed patterns just an average of indivs
- Guy (1980:30): “speakers have essentially identical norms [linguistic constraints] for final /t,d/ deletion, and...one only needs to obtain large amounts of data to demonstrate this fact”
- Tamminga et al. (2016:307): speech community is “a group of people who share the same constraints on, and social evaluation of, intraspeaker variation”



Guy (1980)

- Compared constraints on coronal stop deletion (CSD) among 23 English speakers (19 Philly, 4 NYC)
- Not all speakers had the same constraints. However...
 - Systematic difference between Philly & NYC in ranking of following pause constraint
 - The more tokens a speaker had, the more their constraints obeyed the group pattern
- Hence SCSC

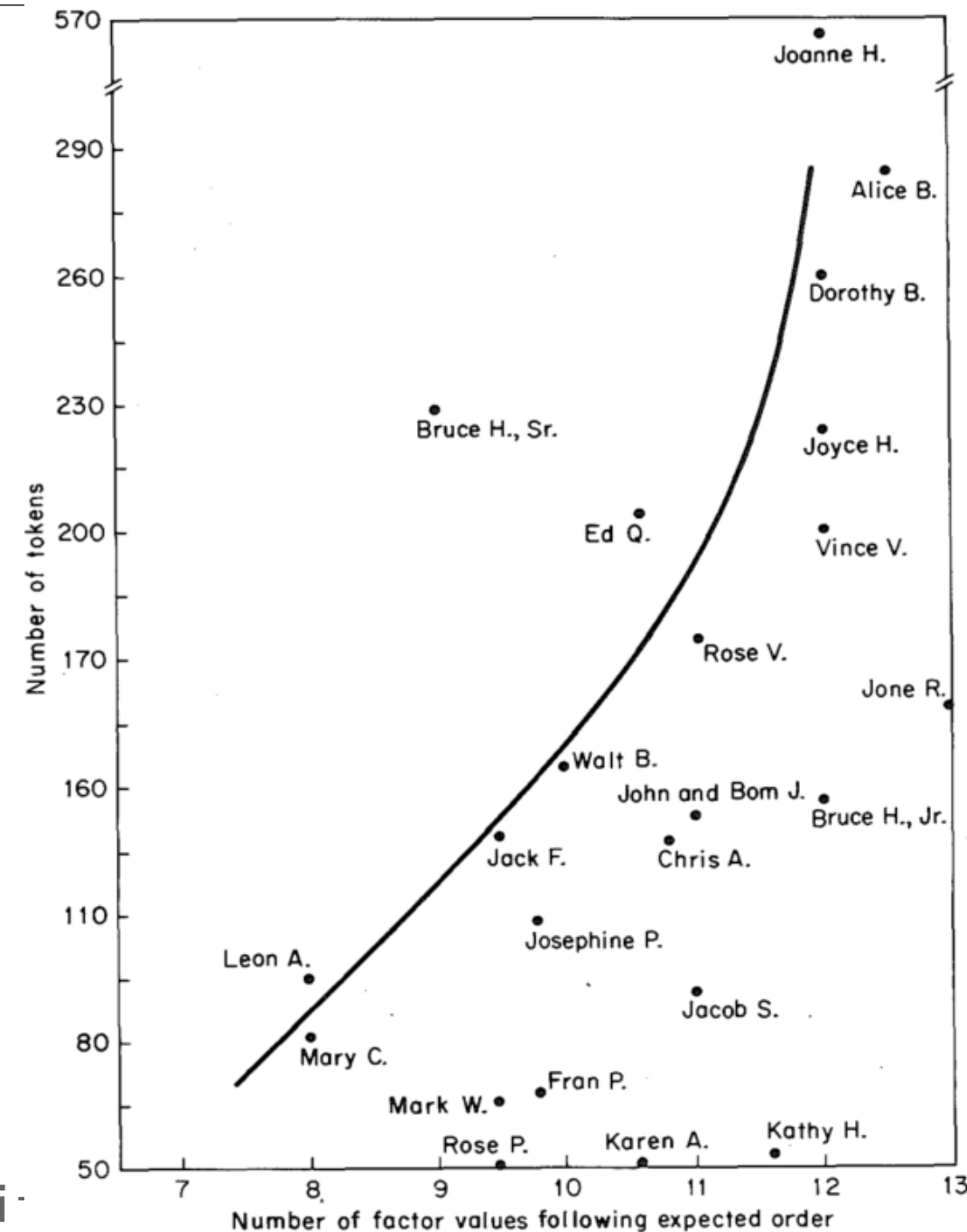


FIGURE 1.4 *Individual accuracy chart.*

The present study

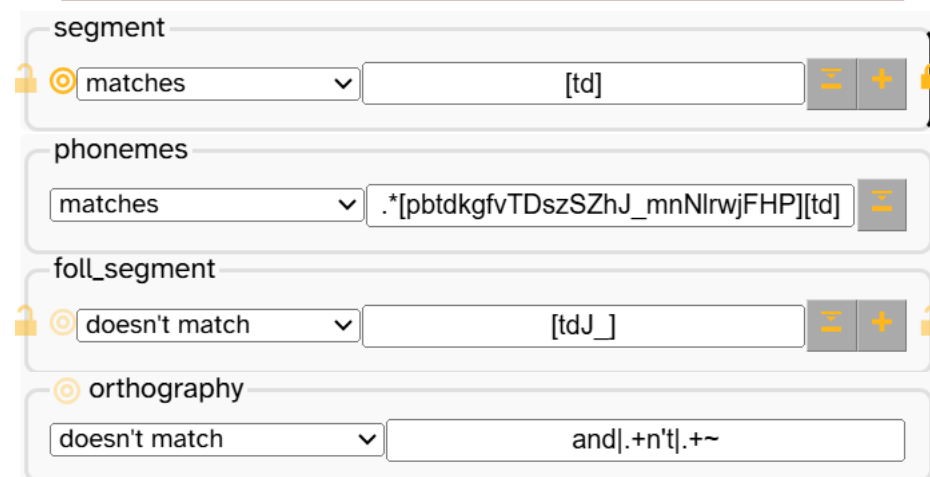
- Our position: It behooves us to perform replications for studies like Guy (1980) that form the cornerstones of our theory and methods
 - And it's easier than ever thanks to open science tools!
- Research questions:
 1. What's the community grammar of coronal stop deletion in Pittsburgh? Is it different for Black vs. white speakers?
 2. To the extent that individuals' grammars deviate from the community, can this be explained as insufficient data?
 - In other words, do Guy's (1980) findings replicate?
Do we find further support for SCSC?



Data

- Data was extracted from the Archive of Pittsburgh Language and Speech (APLS) version 0.4.3 (Villarreal et al. 2026), a publicly accessible corpus of sociolinguistic interviews with Pgh Eng speakers
 - 45+ hours of audio from 40 interviewees (10 Black, 30 white)
 - Annotated with metadata and time-aligned linguistic information
- Variable context:
 - Word-final /t d/
 - Preceded by another consonant
 - Not immediately followed by /t d tʃ dʒ/
 - Not in *and*, *-n't*, or incomplete words

Mapped to regex search in APLS



The screenshot shows a search interface with four sections, each with a dropdown menu and a text input field. The first section, 'segment', has a 'matches' dropdown and '[td]' in the input field. The second section, 'phonemes', has a 'matches' dropdown and '.*[pbtdkgfvTDszSZhJ_mnNlrwjFHP][td]' in the input field. The third section, 'foll_segment', has a 'doesn't match' dropdown and '[tdJ_]' in the input field. The fourth section, 'orthography', has a 'doesn't match' dropdown and 'and|.n't|.+' in the input field. Each section has a lock icon on the left and a plus/minus icon on the right.

Field	Search Type	Regex
segment	matches	[td]
phonemes	matches	.*[pbtdkgfvTDszSZhJ_mnNlrwjFHP][td]
foll_segment	doesn't match	[tdJ_]
orthography	doesn't match	and .n't .+

Analysis

- Used `CodeTokens.praat` (<https://github.com/djvill/code-tokens-dot-praat/>) for rapid auditory coding
 - If auditory evidence of coronal/glottal closure: “retained”; else: “deleted”
- Recreated Guy’s factor groups by extracting phonological environment and morphological information from APLS
- 11.8K CSD tokens (median 270/speaker), 55% deleted
 - Guy (1980): 3,629 tokens (median 133/speaker)
- Ran `glm(deletion ~ grammatical + foll_segment + prec_segment)` separately for each speaker in R 4.5.2
 - Transformed model coefs to uncentered factor weights



Community grammar: Methods

- Perform pairwise comparisons between factor weights, with each speaker getting one “vote” toward community pattern
 - That is, *not* the same as a single model (where speakers with more tokens get more votes)

	M	A	P
M	—	III II	
A	 II	—	II I
P	 III III	 I III	—

Guy (1980:16-17)

FIGURE 1.2 *Pairwise comparisons of factors.*

<https://apls.pitt.edu>



Community grammar

Community	Grammatical	Foll segment	Prec segment
Philly (n=19)	Monomorpheme > Ambiguous > Past tense	Other consonant > Liquid > Glide > Vowel > Pause	Sibilant > Stop > Nasal > Lateral > Other fricative
NYC (n=4)	”	... > Pause > Vowel	(it's complicated— lots of ties)
White Pgh (n=30)	”	” (NYC pattern)	Nasal > Sibilant > Stop > Other fricative > Lateral
Black Pgh (n=10)	”	”	... > Lateral > Other fricative

Community grammar: Summary

- Black and white Pittsburghers have the same constraint hierarchy as one another, save for a low-ranked pair of preceding-segment factors
- Compared to Guy's Philly and NYC speakers:
 - Grammatical is the same
 - Following segment matches NYC (Pause > Vowel)
 - In Pittsburgh, preceding nasal outranks other preceding environments (cf. 3rd-ranked in Philly/NYC)
- But not all individuals follow the group pattern



Individuals vs. groups

In each cell:

- Leftmost # = speakers following group pattern
- Top = Philly/white Pgh
Bottom = NYC/Black Pgh
- Factor pairs vary between **high agreement...**
- ...and **deviations**
- Related to n ?

TABLE 1.4
Pairwise Comparison of Constraints on -t,-d Deletion
for 19 Philadelphians and 4 New Yorkers^a

M			
10-5-3 2-0-2	A		
18-1-0 4-0-0	16-0-2 3-0-1	P	
K			
12-1-4 2-0-2	U		
15-1-3 2-0-2	11-1-5 2-0-2	G	
16-1-2 4-0-0	14-0-3 3-0-1	14-0-5 3-0-1	V
19-0-0 2-0-2	15-0-2 2-0-2	17-0-1 2-0-2	18-0-1 0-0-4
Q			

<https://apls.pitt.edu>

M mono-morpheme			
20-0-10 7-0-3	A ambiguous		
29-0-1 10-0-0	20-0-10 8-0-2	P past	

K consonant			
20-0-7 10-0-0	U liquid		
26-1-2 9-0-1	17-0-10 6-0-4	G glide	
29-0-1 10-0-0	25-0-2 10-0-0	22-0-7 6-0-4	Q pause
30-0-0 10-0-0	24-0-3 10-0-0	25-0-4 10-0-0	20-0-10 7-0-3 V vowel

Individuals vs. groups

Each symbol = 1 speaker

| = factor weights in expected order
o = in opposite order
- = equal

- “For this particular rule, an acceptable, reproducible level of accuracy is not obtained until each factor has **at least 10 tokens** representing it” (p. 20)

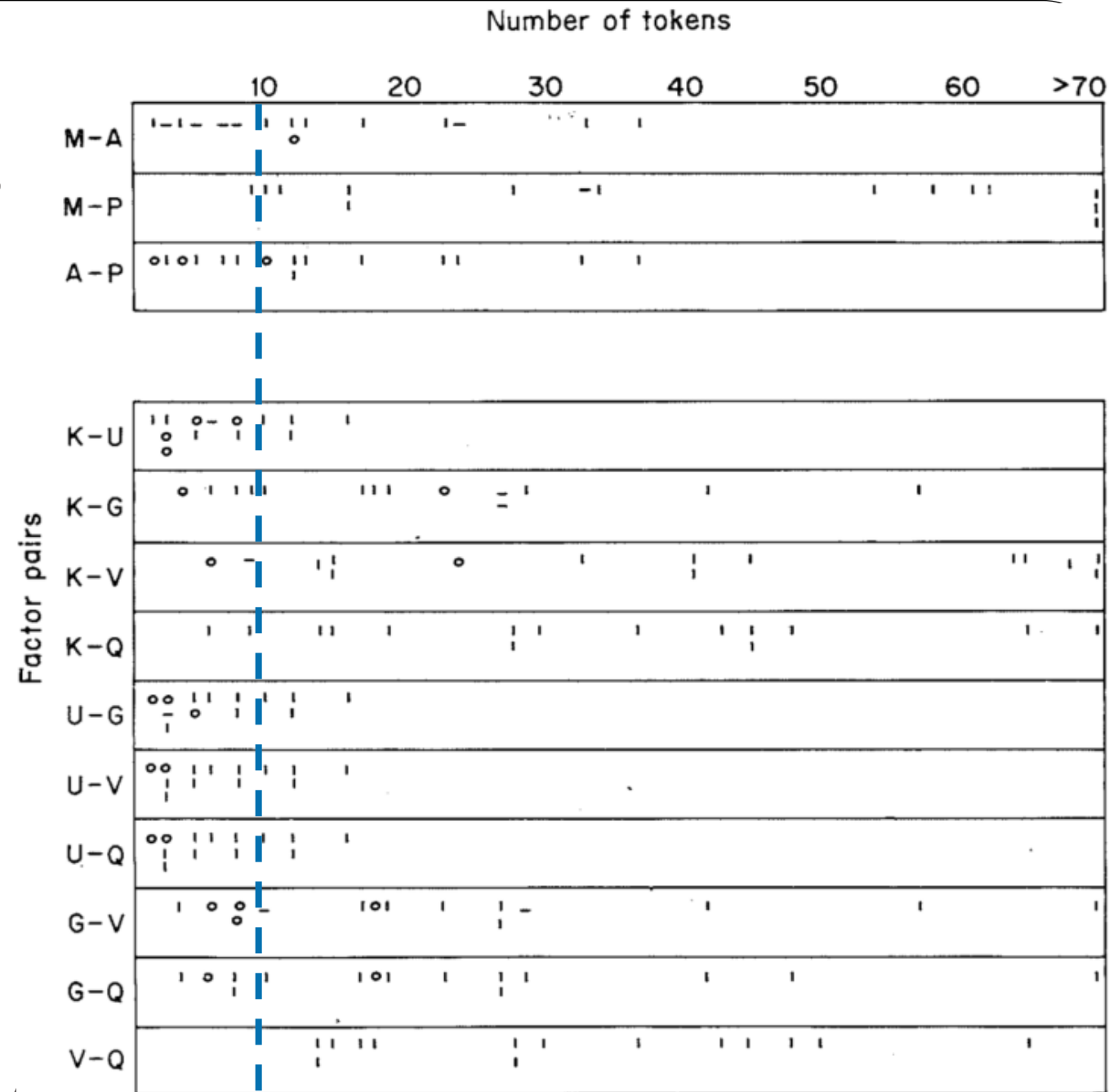


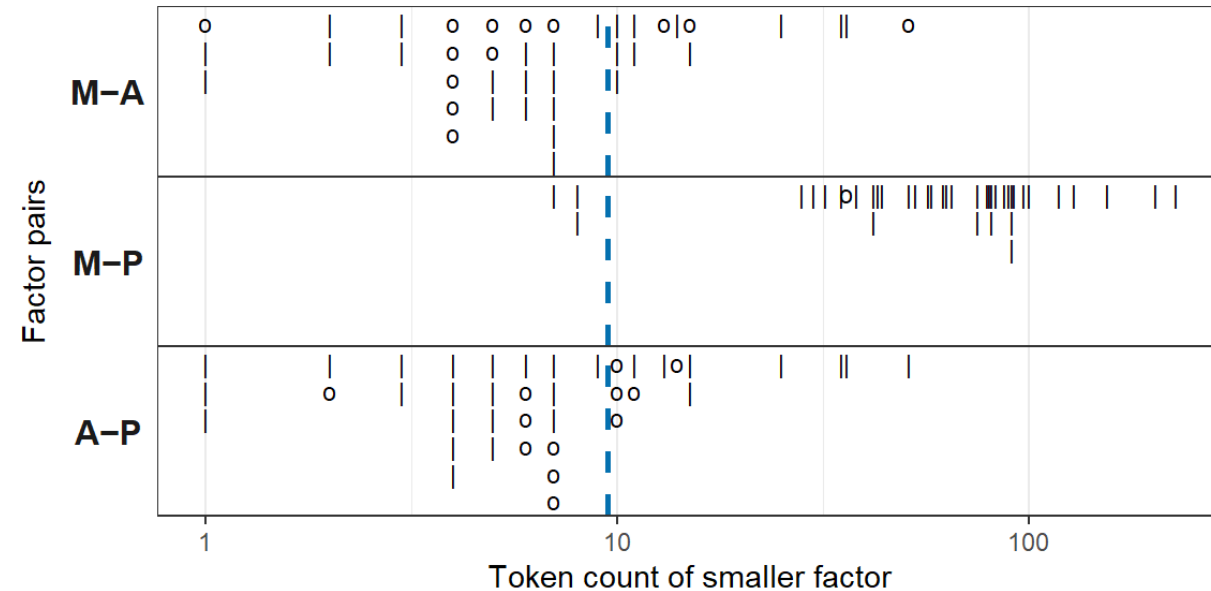
FIGURE 1.3 Pairwise comparisons of factors by data quantity.

Individuals vs. groups

Each symbol = 1 speaker

| = factor weights in expected order
o = in opposite order
— = equal

- “For this particular rule, an acceptable, reproducible level of accuracy is not obtained until each factor has **at least 10 tokens** representing it” (p. 20)
- Grammatical? 

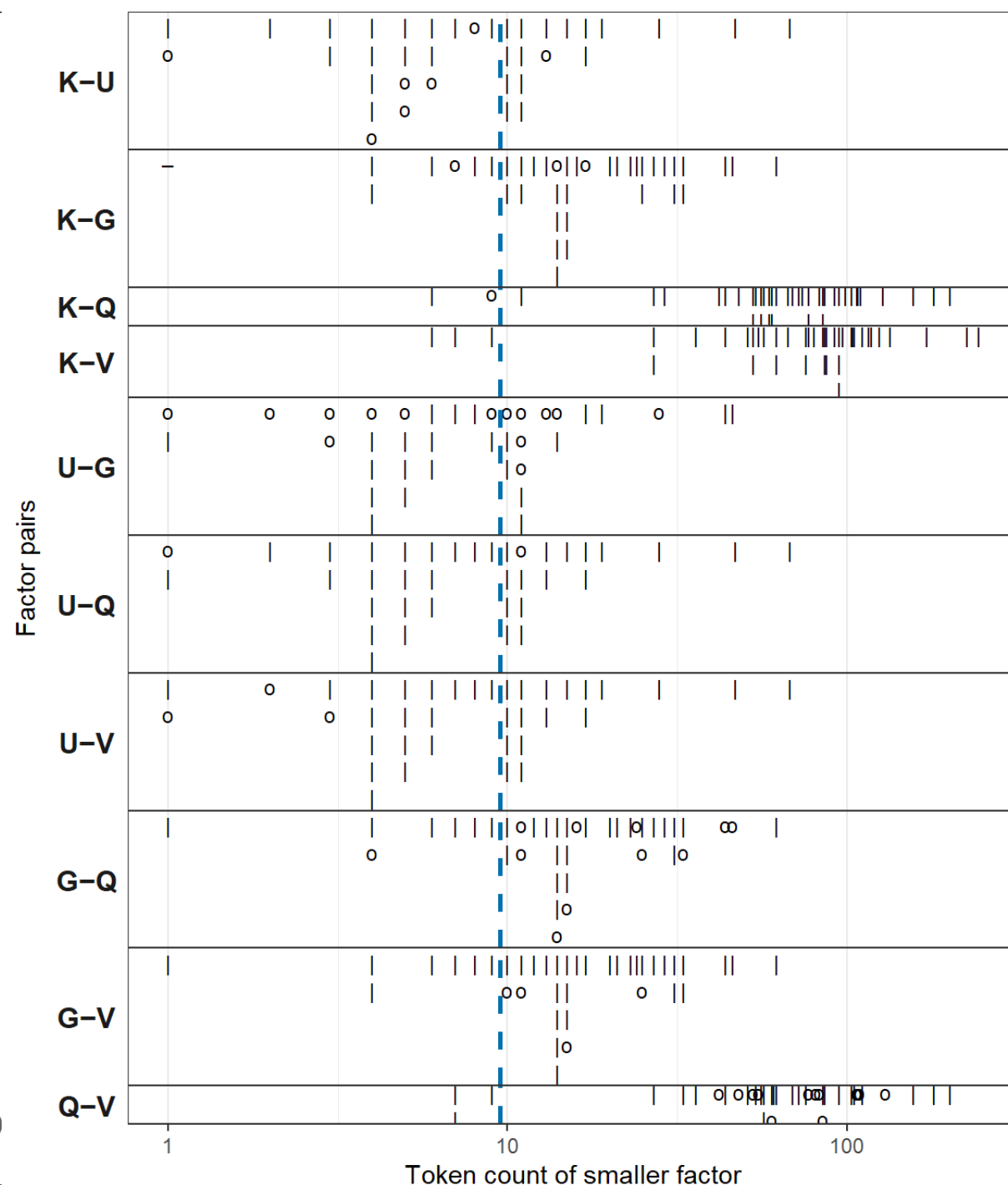


Individuals vs. groups

Each symbol = 1 speaker

| = factor weights in expected order
o = in opposite order
— = equal

- “For this particular rule, an acceptable, reproducible level of accuracy is not obtained until each factor has **at least 10 tokens** representing it” (p. 20)
- Grammatical? ✓
- Foll segment? ✗



Individuals vs. groups

- For each speaker, count the number of factor pairs deviating from the group's ordering, excluding:
 - weights within 0.03 of one another
 - missing weights (i.e., no tokens for one of the factors)
 - preceding fricative vs. lateral (F vs. L)
 - preceding stop vs. nasal/fricative/lateral (\$ vs. N/F/L)
- $\text{NumExpected} = 13 - \text{NumDeviant}$

Guy (1980:22, fn. 11)

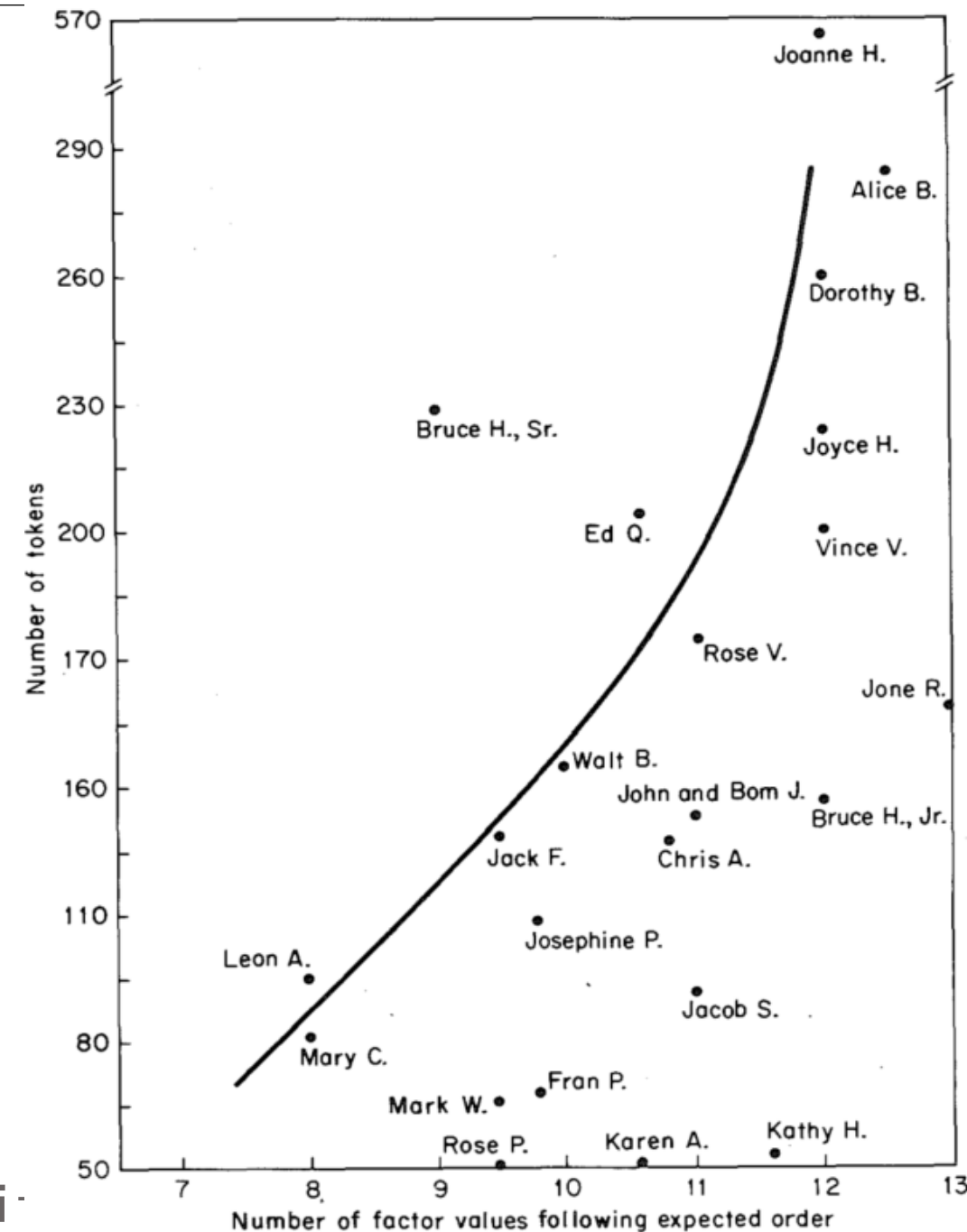


FIGURE 1.4 Individual accuracy chart.

Individuals vs. groups

- In Pgh, no evidence of a relationship between sample size and adherence to group pattern
- Spearman's $\rho = .2151$ ($p = .18$)

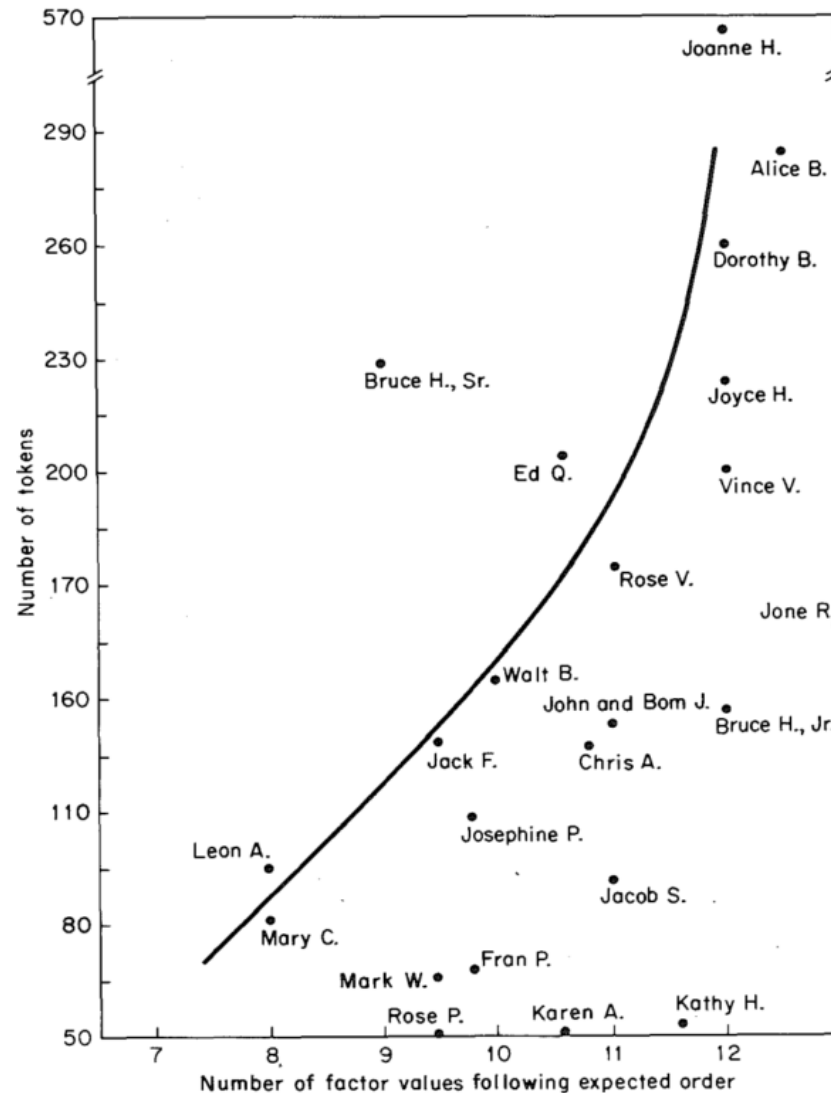
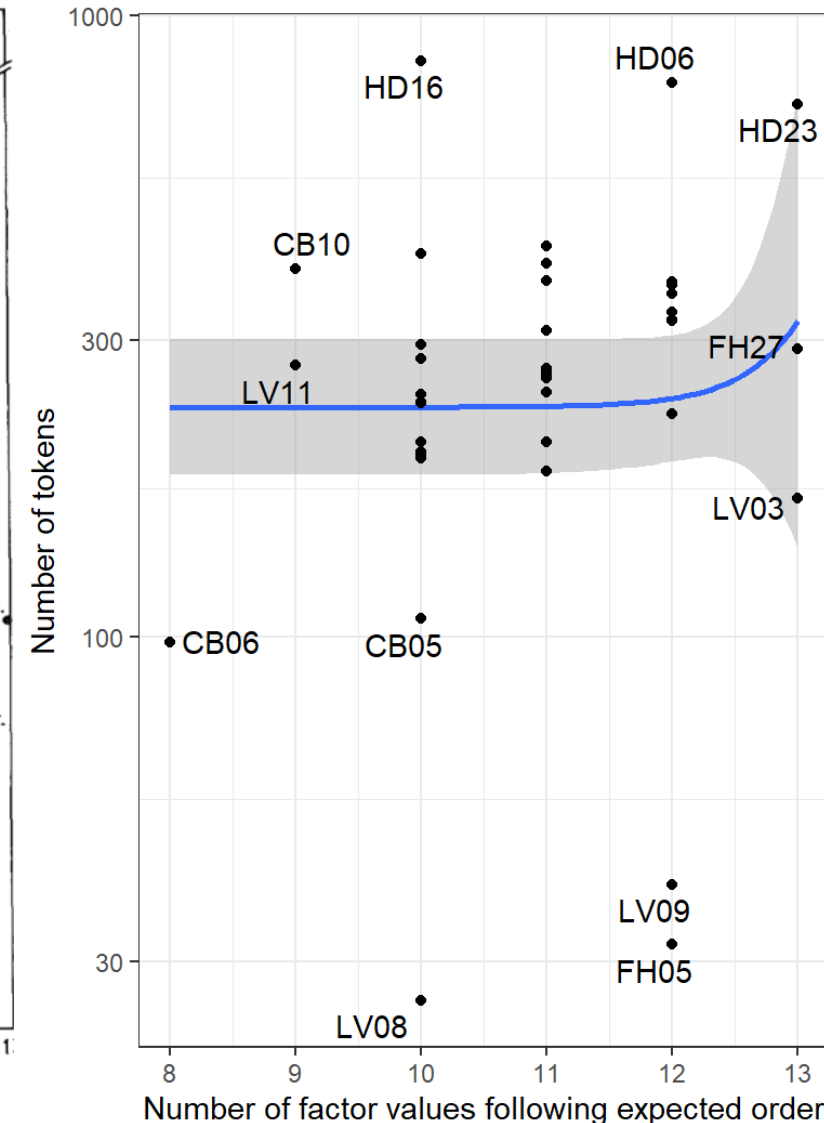
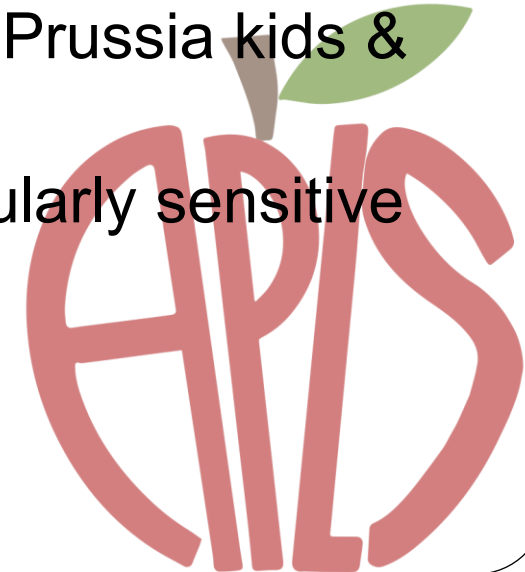


FIGURE 1.4 Individual accuracy chart.



Discussion

- Topline results from Guy (1980) largely didn't replicate
 - Individual deviations from group pattern were attributable to insufficient quantities of data ($n < 10$) for grammatical but not following segment
 - This is in spite of more data/speaker, which should suppress deviations
- Guy's argument also rests on (numerical) factor weights and systematic patterns in deviations—we've left this to future study
 - Guy finds that speakers with $A \geq M$ cluster around King of Prussia kids & working-class adults ("high-A lects")
 - Then again, in Philly/NYC as in Pittsburgh, $A \geq M$ is particularly sensitive to sample size. More work is needed!



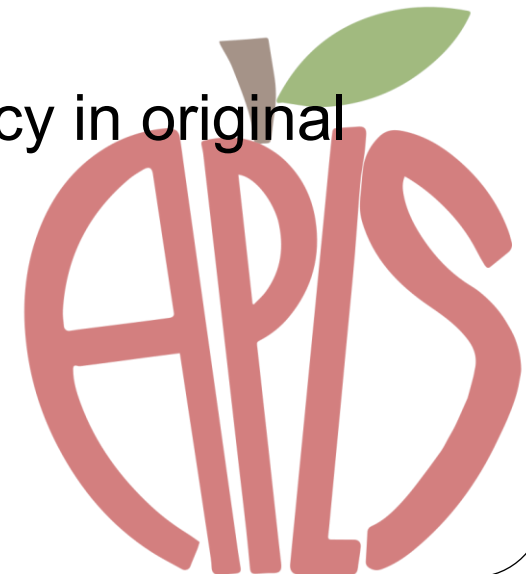
Discussion

- Using APLS substantially improved efficiency and reproducibility
 - Searching for tokens took mere minutes & was systematic
 - Downloaded following segment, preceding segment, morphemes, demographics directly from the corpus
 - Only needed to manually code variant (super-fast thanks to `CodeTokens.praat`) and derive constraints from corpus data
- But it's not perfect.
For example, errors induced by force-alignment:
 - Tokens in non-aligned utterances excluded
 - Some false positives/negatives for following pauses



Conclusions

- Our findings suggest that SCSC as currently understood and applied may be too strong, too blunt an instrument, or both
 - A weaker SCSC would be consistent with Guy (1980). Titled “Variation in the **group** and in the individual”, not **(speech) community!**
- This study demonstrates a key benefit of publicly accessible sociolinguistic corpora like APLS: Being able to perform replications (and have those replications be auditable by readers)
 - Replication also facilitated by methodological transparency in original publications!





These slides

APLS signup
form



Thanks! Questions?

Thanks to Scott Kiesling, Barbara Johnstone, Robert Fromont, members of the Computational Sociolinguistics Lab past and present, and—most of all—the speakers who have shared their voices with us

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